

Introduction to Network Forensics & Security Infrastructure

Special Focus: Power Grid Intrusion Detection

Dr. Pruthvish Rajesh Rajput

School of Cyber Security and Digital Forensics,
National Forensics Sciences University, Gandhinagar

August 10, 2026



Lecture Agenda

- **Fundamentals:** Network Forensics vs. Endpoint Forensics
- **Infrastructure:** Traffic Capture (TAPs, SPAN, Proxies, VPN Tunnels)
- **The Feature Landscape:** Flow-Level (CICFlowMeter) vs. Encrypted Traffic Analysis
- **Securing the Perimeter:** The Role of VPNs, Proxies, and Firewalls
- **The Paradox:** Supervised RF (100%) vs. Unsupervised (0%) on Tunnelled Traffic
- **Live Demo:** Real-Time PCAP Tunnel Analysis

1. Introduction to Network Forensics

The Fundamental Core

Network forensics is the continuous capture, recording, and analysis of network traffic to investigate security breaches and discover attack paths.

- **Endpoint Forensics (The Crime Scene)**
- **Network Forensics (The Highway Cameras)**

Endpoint Forensics (The Crime Scene)

- Analyzes Remote Terminal Unit (RTU) logs, Human Machine Interface (HMI) databases, and RAM.
- Vulnerability: Advanced industrial malware (e.g., Industroyer) can alter or wipe local operational logs.
 - Industroyer¹ (also called CrashOverride) is a highly sophisticated, destructive malware framework explicitly designed to target and disrupt electrical grids and other critical infrastructure.
 - It can directly manipulate industrial substation switches and circuit breakers and typically ends its attack sequence by deploying a data wiper to destroy system hard drives.
 - The malware is widely believed to have been used in the December 17, 2016, cyberattack on Ukraine's power grid.

¹<https://www.welivesecurity.com/2017/06/12/industroyer-biggest-threat-industrial-control-systems-since-stuxnet/>

Network Forensics (The Highway Cameras)

- Analyzes traffic in transit. Investigates the packets in transit inside the network.
 - Source-Destination analysis
 - Protocol-wise traffic analysis
 - Time-based analysis
 - Payload analysis
- Hard Truth: Attackers cannot hide physical packets sent on the wire over industrial networks.

2. Security Network Infrastructure Overview

To extract forensic evidence without altering critical substation runtime execution, sensors must intercept data cleanly.

The Visibility Triad

An effective industrial network forensic deployment relies on three interception mechanisms:

- **Network TAPs:** Hardware-level passive splitting (Best for 100% capture integrity in critical links).
- **SPAN / Port Mirroring:** Switch-level software copying (Best for rapid diagnostic deployment).
- **Substation Bus TAPs:** Passive capture points directly monitoring the process and station buses.

2.1 Network TAPs (Test Access Points)

A physical hardware device inserted directly between two network nodes (e.g., an industrial switch and a gateway/RTU).

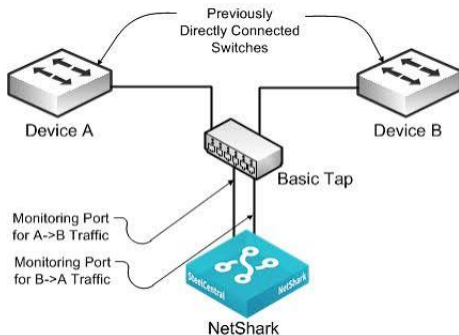


Figure: Cyber Security Concepts



Figure: Cyber Security Concepts

2.1 Network TAPs (Test Access Points)

How It Works:

- It breaks the physical cable connection.
- Internal passive circuits split the signal, sending an isolated copy to the forensic monitor.
- Completely invisible to the network (no IP or MAC address to target).

Forensic Trade-offs:

- **Pro:** Captures 100% of packets, including corrupted/malformed frames and CRC alignment errors.
- **Con:** Introduces a brief communication drop during initial physical installation.

2.2 SPAN Ports (Switched Port Analyzer)

A software-defined configuration on a managed industrial ethernet switch.

How It Works:

- The switch ASIC/CPU copies traffic flowing through designated critical ports.
- It mirrors those copies out of a dedicated "Forensic Monitor" port.

Forensic Trade-offs:

- **Pro:** Remotely configurable; zero hardware footprint overhead inside the cabinet.
- **Con:** Under high load or a network-flooding DoS, the switch CPU prioritizes routing over mirroring. **It will silently drop packets**, destroying forensic sequences.

2.2 SPAN Ports (Switched Port Analyzer)

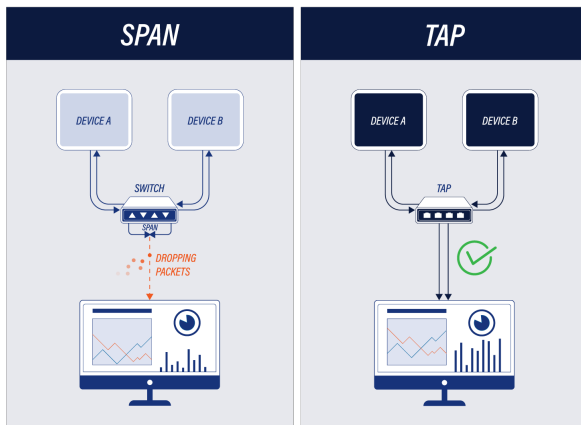


Figure: Cyber Security Concepts

2.3 Securing Infrastructure: Proxies vs. VPNs

How traffic visibility changes when passing through boundary security devices:

Proxies (Application Level)

- Intercepts traffic at Layer 7 (HTTP/SOCKS).
- Terminates client connection; originates a new request.
- **Forensic Impact:** Hides internal Source IP, but allows deep packet payload inspection/filtering.

VPN Tunnels (Network Level)

- Encrypts traffic at Layer 3/4 (IPsec/OpenVPN/WireGuard).
- Wraps entire packet in an encrypted outer header.
- **Forensic Impact:** Blinds deep packet inspection (DPI); forces reliance on flow metadata.

3. Traffic Feature Extraction: CICFlowMeter in Action

When traffic is encrypted via VPNs or routed through proxies, payload inspection fails. We rely on **Network Flow Analytics**:

Flow Feature Engineering (e.g., CICFlowMeter)

Converts raw PCAP streams into bidirectional flow summaries with over 80 features:

- **Time-Series/Pacing**: Inter-Arrival Time (IAT Mean, Std-Dev, Max, Min).
- **Directional Ratios**: Forward vs. Backward packet count and byte ratios.
- **Burst Behaviors**: Active Time vs. Idle Time distributions.

*Core Concept: Even when payload bytes are encrypted by a VPN, the **pacing and shape** of the traffic can reveal proxy usage or malicious tunneling.*

3.1 Standard PCAP Analysis (The Traditional Way)

Analysts open a Packet Capture (PCAP) file using tools like Wireshark or 'tcpdump'.

- **The Core Metric:** Manual inspection of protocol fields (e.g., verifying 'stNum' or 'sqNum' within a GOOSE PDU block).
- **The Process:** Reassembling payload data bytes to evaluate raw hex strings.
- **The Forensic Limit:** Dependent on human observation. If an adversary injects a malicious command inside a high-frequency stream of 500,000 cyclic keep-alives, a human analyst will pass right over it.

3.2 Traditional Flow Analysis (The Network Auditor)

Traditional network models (NetFlow², IPFIX³) compress raw traffic by dropping payloads entirely.

- **The 5-Tuple Keys:** Traffic is grouped strictly if packets share identical:
 - *Source IP, Destination IP, Source Port, Destination Port, Protocol.*
- **The Output:** A structural summary window containing counters: Total Packet Count, Total Byte Count, Duration.
- **The Industrial Limit:** Highly predictable, deterministic cyclic telemetry looks identical in basic NetFlow. Anomalies remain hidden if packet sizes match typical heartbeat patterns.

²<https://www.ibm.com/think/topics/netflow>

³<https://www.ibm.com/docs/en/npi/1.3.1?topic=insight-ipfix-overview>

3.3 Feature Engineering in Literature: Network vs. Semantic

To feed raw substation PCAPs into Machine Learning models, the literature utilizes two distinct feature extraction paradigms:

1. The Network Flow Approach (CICFlowMeter^a)

^a<https://www.unb.ca/cic/research/applications.html>

Converts PCAPs into bidirectional IP flows, extracting **80+ statistical features** (e.g., flow duration, forward/backward packet pacing, network-layer Inter-Arrival Time [IAT]).

Captures *traffic behavior*.

2. The Industrial Abstraction Approach (IPAL)

An open-source framework that strips network overhead to parse raw industrial payloads (GOOSE, Modbus, DNP3). Standardizes variables like state transition counters (*stNum/sqNum*). Captures *operational semantics*.

Preview: The Multi-Dimensional Forensic Demo

We will ingest real substation traffic traces from the **PowerDuck dataset** and process it through our multi-tiered pipeline:

Pipeline Tier 1: Behavioral Metrics

Extracting macro-network stats via **CICFlowMeter** to build our statistical baseline (where our unsupervised model will struggle).

Pipeline Tier 2: State Semantics

Parsing micro-protocol fields via **IPAL** to feed our supervised Random Forest, decoding the true intent of the traffic.

4. The Forensic Ground Truth: The PowerDuck Dataset

To evaluate our IDS, we move away from synthetic simulations and utilize a high-fidelity, real-world baseline.

- **Source:** Developed by Zemanek et al. (2022), *PowerDuck: A GOOSE Data Set of Cyberattacks in Substations*.
- **Environment:** Captured from a **physical power substation testbed** containing genuine hardware latency, network jitter, and background noise.
- **Protocol Focus:** High-speed **IEC 61850 GOOSE** (Generic Object Oriented Substation Events) multicast layers used for real-time protection and control.
- **Abstraction:** Normal and malicious streams are standardized using the **Industrial Protocol Abstraction Layer (IPAL)** format for streamlined feature parsing.

5. Visibility Comparison: Standard vs. Encrypted/Proxied Traffic

Feature	Standard Cleartext Traffic	VPN / Proxied Traffic
Visibility	Full (Headers + Payload)	Partial (Encrypted Outer Header Only)
Attribution	Direct Source/Dest IP	Masked by Proxy/VPN Gateway
Analysis Method	Deep Packet Inspection (DPI)	Flow Metadata (CICFlowMeter IAT)
Evasion Metric	String Obfuscation	Encrypted Tunneling / Port Flipping

6. The Detection Paradox in Tunnelled Anomaly Detection

Testing models on PCAP traces containing proxy/VPN tunnelled attacks reveals a stark performance gap:

Supervised Random Forest

- **Accuracy: 100%.**
- **Why:** Successfully maps subtle combinations of flow features (e.g., specific packet size ratios + IAT signatures of known proxy tools).
- **Explainability:** Validated via **SHAP Analysis**.

Unsupervised Statistical Model

- **Detection Rate: 0%.**
- **Why:** Trained on normal web/VPN browsing. Attackers disguise command-and-control (C2) traffic inside normal HTTPS/VPN flow profiles.
- **Failure:** Statistical distance cannot distinguish malicious tunnelled data from legitimate streaming/browsing.

Unsupervised Statistical Model

- **Performance: 0% Detection Rate** (Total Failure).
- **Why it fails:** Trained strictly on normal steady-states. Attackers execute "semantic manipulation" using perfectly legal thresholds and packet sizes.
- **The Forensic Lesson:** Pure statistical distance and behavioral baselines cannot detect attacks that operate completely within normal physical boundaries.

Transition to Live Demonstration

Telemetry Focus: **PowerDuck GOOSE Core Stream Assessment**

We will now execute our Python forensic pipeline via IPAL, comparing the 100% accurate Supervised RF model (with SHAP feature plotting) against the blind spots of Unsupervised Statistical profiling.

7. Explaining 100% Accuracy (SHAP)

How does the Random Forest achieve 100%? It uncovers the hidden logic of the attacker.

- **The Smoking Gun:** SHAP analysis reveals that `stNum` and `sqNum` are the highest contributors to the model's decision.
- **The Discovery:** Attacker frames often reset *sqNum* incorrectly or use a mismatched *stNum* for the current measurement values.
- **The Reality Check:** This 100% is a "Signature Trap"-the model is brilliant at catching *known* logic violations but might fail on novel ones.